Block practice · Pixels, Rail Fence, hardware devices, environment and the law

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Hardware devices: input, output and how they connect

Protecting software by law: copyright, patents, trademarks and licences

Technology, the environment and health

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Practice workbook · Hardware devices: input, output and how they connect

Input and output devices, the role of hardware versus software, and the direction of communication between peripherals and the CPU.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-02

1. Question 5 · part b.i · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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1. Question 5 · part b.i · 3 marks

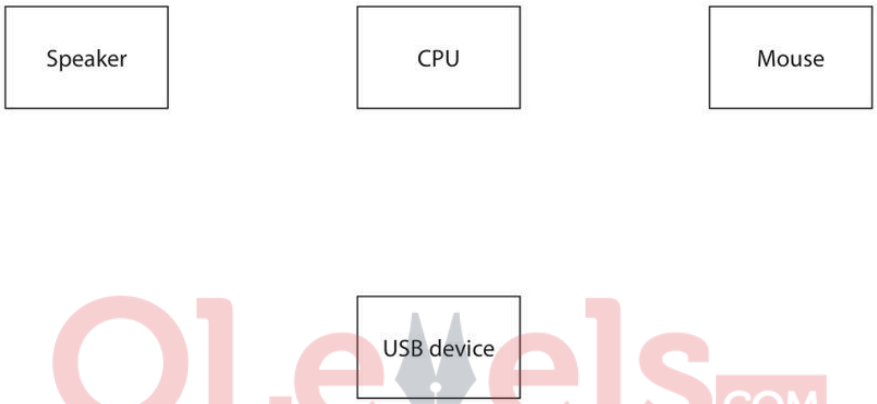
2023 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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P 7 2 9 3 7 A 0 1 5 2 4

15

Turn over

1. Question 5 · part b.i · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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1. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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2. Question 3 · part a · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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2. Question 3 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part d · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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3. Question 3 · part d · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

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(d) Zakir uses many devices.

Complete the table by adding **one** tick (✓) in each row to show whether the device is an input device, an output device or both.

(2)

Device	Input	Output	Both
Headset			
Mouse			
Printer			
Touch screen			

(e) Zakir needs to protect his laptop against malware.

Give **two** types of anti-malware software he could install.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

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Working and answer

4. Question 2 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

.....

2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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P 7 8 6 3 7 A 0 5 2 0

4. Question 2 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Protecting software by law: copyright, patents, trademarks and licences

Legal protections for software and systems (copyright, patents, trademarks, licences) and how licensing controls copying and distribution.

2 question parts: 1 worked-example retry and 1 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-16

1. Question 6 · part d · 6 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

2 .....

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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1. Question 6 · part d · 6 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

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(c) Akiko is also working on a burglar alarm system.

The alarm must only be triggered if:

- the alarm (A) is set
- it is dark outside (O)
- a window (W) or a door (D) has been opened.

Construct a Boolean expression, using AND, OR and NOT with the letters A, O, W, and D to show the conditions that will trigger the alarm.

(4)

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1. Question 6 · part d · 6 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

(d) Akiko wants to know how the software she writes can be protected.

Discuss how the software can be protected from illegal copying and/or illegal distribution.

You should consider:

- intellectual property
- licensing.

(6)

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(Total for Question 6 = 15 marks)

1. Question 6 · part d · 6 marks · Working and answer

Working and answer

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2. Question 5 · part d · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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5 Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

P AND (Q OR R)

You may not need to use all rows.

(3)

P	Q	R	Q OR R	P AND (Q OR R)

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2. Question 5 · part d · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

(2)

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

(2)

1 .....

2 .....

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2. Question 5 · part d · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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(d) The team needs to protect its intellectual property rights over the systems that it develops.

Give **three** types of legal protection the team could use.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(Total for Question 5 = 10 marks)

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Working and answer

Practice workbook · Technology, the environment and health

Energy use, resource extraction, e-waste and actions to reduce them, plus health effects/benefits of technology such as wearables.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-14

1. Question 5 · part c · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

1. Question 5 · part c · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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DO NOT WRITE IN THIS AREA

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P 7 8 6 3 7 A 0 1 4 2 0

1. Question 5 · part c · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c only.

(c) Wearable technology, such as smartwatches, has become very popular with many groups of people including athletes.

Discuss how the use of wearable technology can be a benefit to health.

(6)

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(Total for Question 5 = 16 marks)

1. Question 5 · part c · 6 marks · Working and answer

Working and answer

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2. Question 6 · part c · 6 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

(i) Explain **one** difference between DNA computing and traditional computing.

(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

(b) Ethically, everyone should be able to access the benefits of technology.

Give **two** reasons why this may not be possible.

(2)

1

2

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16

P 6 6 4 8 9 A 0 1 6 2 0

2. Question 6 · part c · 6 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

(c) Widespread use of computer technology has a negative environmental impact.

Discuss the negative environmental impact of using computers and actions that could be taken to reduce it.

(6)

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(Total for Question 6 = 11 marks)

2. Question 6 · part c · 6 marks · Working and answer

Working and answer

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3. Question 3 · part e.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part e.i only.

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3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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3. Question 3 · part e.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part e.i only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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3. Question 3 · part e.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part e.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part e.ii only.

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(2)

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Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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4. Question 3 · part e.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part e.ii only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
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- ☒ **C** 4.1 GHz
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(1)

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(2)

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(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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4. Question 3 · part e.ii · 2 marks · Working and answer

Working and answer

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